

Delivery Company System

Al-ghala Al Kalbani 24-0309

Dana Abu Quta 24-0768

Lama Al Maharibi 24-0747

Zainab Abdul Redha 24-0693

Maryah Al Mujaini 24-0542



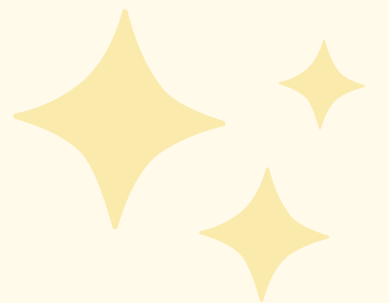
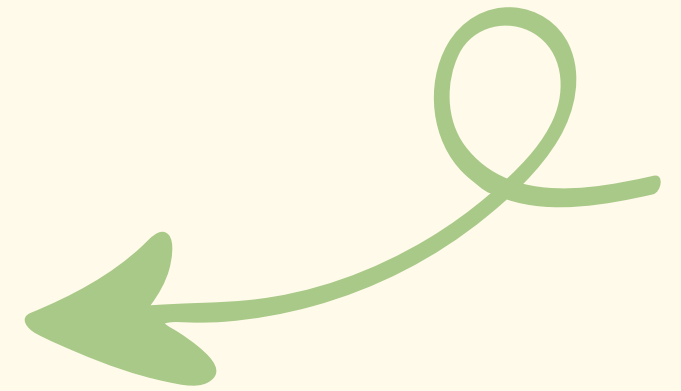
System Overview



Purpose: Manage customer orders, assign drivers, and track deliveries

Features:

- Manage customer data
- Create and process orders
- Assign drivers dynamically
- Track delivery status
- Estimate delivery time (ETA)



Main classes

01

Customer

02

Order

03

Driver

04

Company

Customer Class

Role: Represents the person placing an order (stores customer information used in orders.)

Main data:

- Name
- Address
- phone number

Concepts used :

- Encapsulation (private data, public getters/setters)
- Constructors/Destructors for initialization
- Exception handling for invalid contact info

Relation: Composed inside Order (an order has-a customer).



Order class

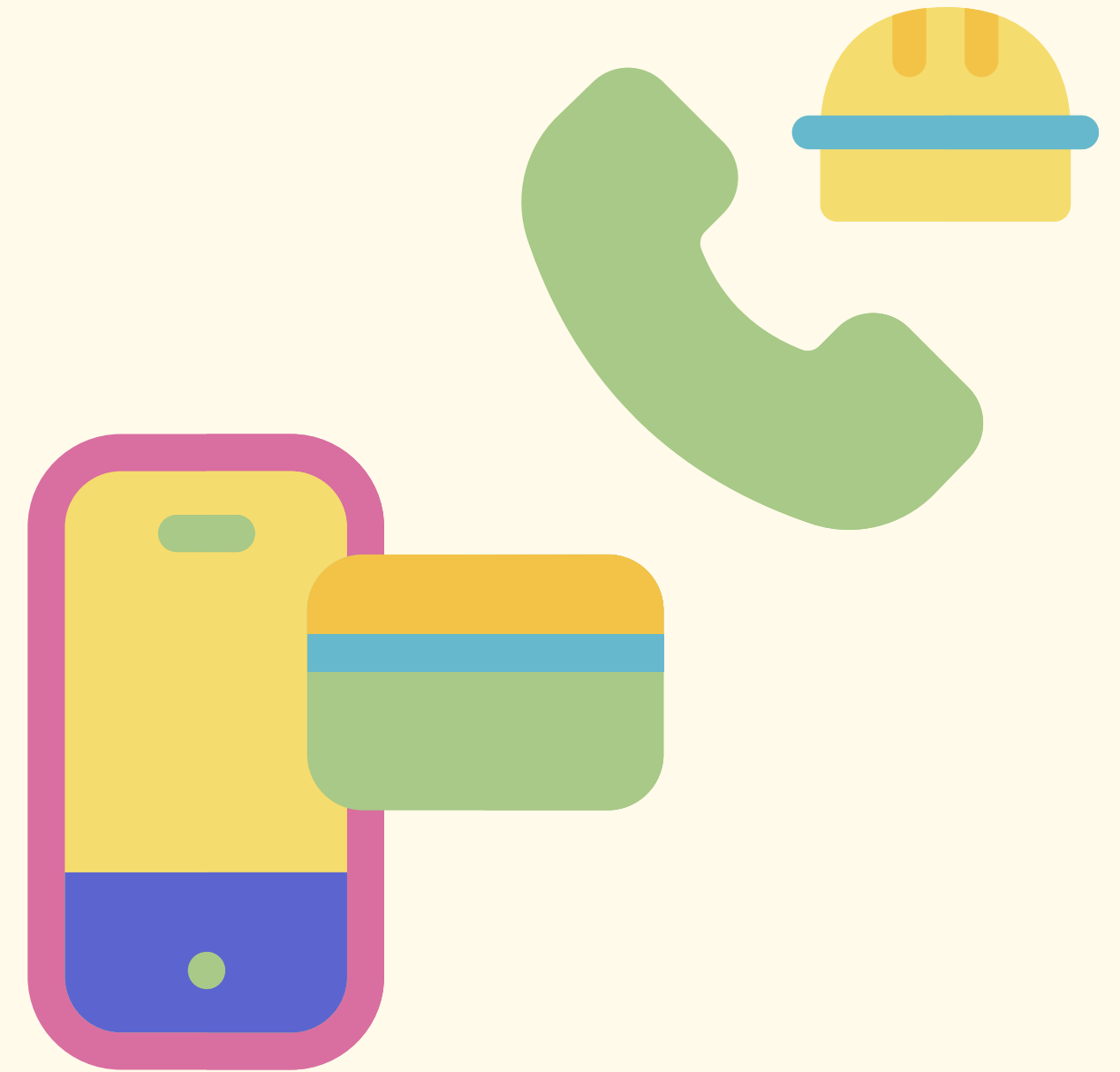
Role: Handles all data related to a delivery order.

Attributes:

- orderID
- itemDescription
- status
- driver

Concepts demonstrated:

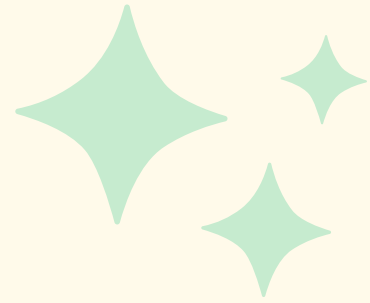
- Composition (contains a customer object)
- Operator overloading (<< for printing and == for comparing orders)
- Exception handling (invalid status or no driver)
-
- **Relation:** Aggregates a Driver (pointer to driver)



Responsibility: Represents delivery staff.

Key Data:

- Driver ID
- name
- vehicle type



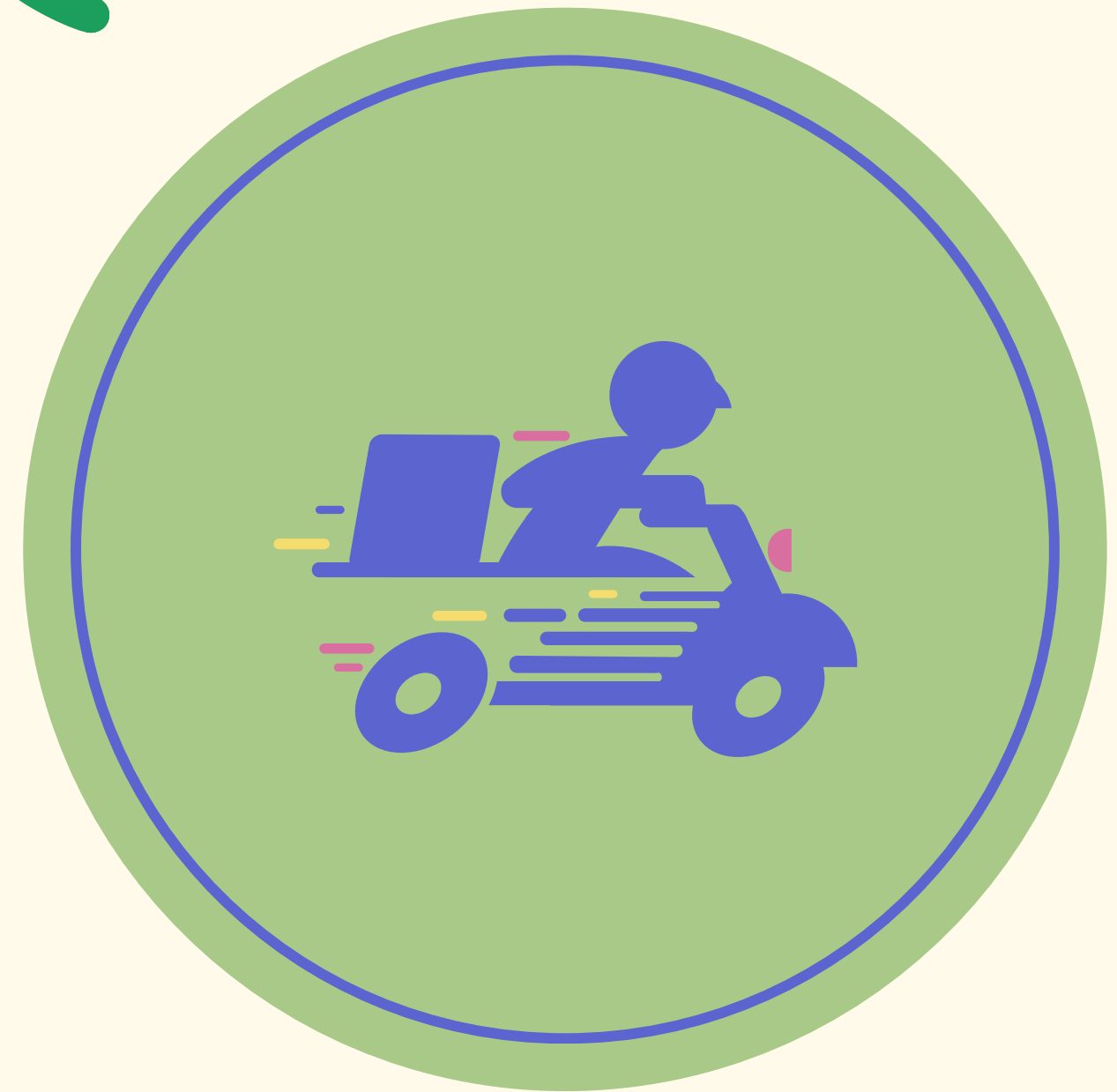
Concepts Demonstrated:

- Inheritance (CarDriver, BikeDriver subclasses)
- Polymorphism (virtual calculateETA() overridden in subclasses)

Relationship:

- Assigned to Orders
- Managed by Company

Driver class



Company Class

- **Responsibility:** Central system managing orders, drivers, and budget (control the whole system)
- **Key Data:** Arrays of orders and drivers, budget objects.
- **Concepts Demonstrated:**
 - Composition (has a Budget)
 - Aggregation (manages pointers to orders and drivers)
 - Exception handling (invalid IDs, unavailable drivers, budget overflow)
- **Relationship:** Connects all classes together.

